

ASAD IKRAM

Data Engineer and Computational Social Science Researcher · MSc Data Analytics (Distinction) · Chevening Scholar 2024/25

asad.ikram53@gmail.com · +44 7482 376417 · Lahore, Pakistan · linkedin.com/in/asad-ikram98 · asad-ikram-mc.github.io/portfolio · web-scraping-guide.com

PROFILE

Data engineer with 6+ years building large-scale social media collection and analysis systems, moving into doctoral research on how online platforms shape what we measure of public opinion. My MSc dissertation applied transformer-based sentiment and emotion analysis to 279,000 posts across four platforms; a methods paper is in preparation with my supervisor. I bring production experience in exactly what this project names: platform data scraping (I wrote the public reference on it: web-scraping-guide.com), multimodal and emotion analysis of social media at scale, and engagement prediction.

RESEARCH

MSc Dissertation: Cross-Platform Sentiment Analysis of Public Reaction to UK Economic Policies

University of Portsmouth, 2025 · Distinction (Masters Project mark: 78) · Supervisor: Dr Ella Haig

- Collected and integrated ~279,000 public posts from Twitter/X, Reddit, YouTube and Quora on UK economic policy via official APIs and licensed feeds, under ethics approval and GDPR-compliant procedures.
- Fine-tuned BERT classifier (macro-F1 0.878) against BiLSTM and lexicon baselines; human validation (Cohen's kappa ~0.78); temperature-scaling calibration; McNemar and bootstrap significance testing; chi-square tests of topic differences; Prophet forecasting of the daily sentiment index (MAE ~0.02).
- NRC emotion analysis linking anger and fear to negative policy reactions; interpretability via LIME and integrated gradients.

Paper in preparation, with Dr Ella Haig (University of Portsmouth)

- Treats each platform as a survey mode and applies psychometric measurement tools (generalizability theory, invariance and DIF testing, test equating) to machine-labelled sentiment, separating genuine opinion differences from platform measurement effects. Reviewed by an ICWSM-experienced researcher; arXiv preprint planned late 2026, then ICWSM 2027.

PROFESSIONAL EXPERIENCE

Founder & CTO, ArtemisAI Ltd (trading as ViralData)

Feb 2025 - present · UK company, remote

- Founded and lead a social media intelligence SaaS analysing public Facebook pages; built the full platform: Meta Graph API ingestion of 30+ brand pages daily into a layered lakehouse and Redshift warehouse (111,000+ posts and 3.2M+ comments processed to date), launching Nov/Dec 2026.
- Designed and deployed six production NLP model endpoints on AWS SageMaker: sentiment, emotion, toxicity, topic, language and intent classification of social media text at scale.
- Built multimodal analysis into the product: CLIP visual embeddings of post images and Node2Vec graph embeddings (Pinecone vector database), so image and network context are analysed alongside text - the same text-plus-image lens this project applies to migration narratives.
- Trained an XGBoost engagement-prediction model on 52 features and a crisis (viral storm) detection system for public-page conversations: predicting when online content mobilises audience response.
- Built an LLM analysis layer (RAG pipeline routing GPT-4 and fine-tuned LLaMA) that turns raw social data into analyst-readable narratives; currently auditing Meta API permissions, rate limits and version deprecations for launch.

Data Engineer Consultant, M+C Saatchi Fluency

Dec 2024 - present · UK, remote

- Designed and operate the multi-platform social listening pipeline behind campaign intelligence for UK Government departments, Amazon, Ford, Nike and Reckitt: Sprinklr and Brandwatch feeds plus custom crawlers into AWS (ECS Fargate, Step Functions, Glue, Athena), with SageMaker NLP sentiment models; reports are read by senior government communications teams.
- Tracked narrative shifts, misinformation dynamics, polarising themes and sentiment trends across millions of conversations for national-level public communication campaigns.

- Run the Ford LENS crawler fleet (~70 Scrapy spiders across 44 domains) with CAPTCHA handling, proxy rotation and TLS fingerprint management; 99%+ pipeline uptime with CloudWatch monitoring and freshness dashboards.
- Maintain strict data governance on government-facing datasets: documented filter rules, permanent removal lists, pre-delete backups and weekly refresh cycles.

Senior Data Engineer, Dubizzle Labs (Dubizzle, Bayut, Zameen, OLX)

Feb 2022 - Aug 2024 · Pakistan, onsite

- Led a 3-engineer team owning 500+ regional web scrapers across 15 MENA countries on Kubernetes/EKS, collecting multilingual platform data (English, Arabic, Urdu, Hindi) for market and consumer-behaviour analytics.
- Redesigned the layered data warehouse (RDL, ODL, ADL), improving ETL/ELT throughput 60%; built the competitor-intelligence platform combining scraped data with warehouse analytics used by leadership.
- Promoted twice in two years: Associate Software Engineer to Data Engineer to Senior Data Engineer.

Data Engineer Consultant, Fix.com

May 2023 - present · Canada, remote (part-time)

- Designed Scrapy/ScrapyD pipelines processing 50M+ data points from 200+ e-commerce and forum sources; cut crawl time 40% via distributed Scrapy Cluster tuning.
- Introduced Great Expectations data validation with automated schema-drift detection, reducing bad production data by 70%.

Lead Data Engineer, VendueTech

Jun 2025 - Jul 2025 · Ireland, remote contract

- Led a 3-engineer team building real-time auction data ingestion (Scrapy, Playwright, AWS Batch, Step Functions); migrated infrastructure Azure to AWS and cut data latency 40% with PySpark on EMR Serverless.

Data Engineer Consultant, Prefe

Sep 2024 - Dec 2024 · Italy, remote

- Rebuilt legacy scrapers into a modern Scrapy architecture, reducing failure rate 60%; mentored two junior engineers and established team-wide code review standards.

Freelance Data Engineer, CXG

Dec 2025 - Jan 2026 · Dubai, remote

- Built production crawlers for global luxury fashion brands with 99%+ data completeness against advanced anti-bot protections.

Senior Freelance Data Engineer (Level 2 seller, 300+ projects)

Dec 2018 - Feb 2026 · worldwide

- Delivered 300+ five-star web scraping and data engineering projects across 400+ websites for 50+ international clients, including full-stack data products from collection to warehouse to dashboard.

EDUCATION

MSc Data Analytics, Distinction

University of Portsmouth, UK · 2024 - 2025

Modules incl. Intelligent Data and Text Analytics, Big Data and Machine Learning, Explainable AI, Data Management (80)

BSc Computer Science

FAST-NUCES, Pakistan · 2017 - 2022

AWARDS AND PUBLIC WORK

Chevening Scholar 2024/25: UK FCDO flagship scholarship, selected from 70,000+ applicants across 160 countries.

Author, web-scraping-guide.com: a free public engineering reference on web data collection; 13 sections covering 86 libraries and tools, six anti-bot vendors, legal and ethical constraints, production architectures and data quality.

METHODS AND TOOLS

Quantitative and ML: BERT and transformer fine-tuning, BiLSTM, XGBoost, statistical testing (chi-square, McNemar, bootstrap CIs), calibration, Prophet time-series forecasting, LIME/attribution, NRC and VADER lexicons, CLIP multimodal embeddings.

Data collection: Scrapy, Playwright, Selenium, official platform APIs (Meta Graph, Reddit, YouTube), licensed feeds, proxy and anti-blocking management, multimodal content (text, images, video metadata).

Engineering: Python, PySpark, SQL, AWS (Fargate, Glue, Athena, Redshift, SageMaker), Airflow/Step Functions, Docker/Kubernetes.

Languages: English (fluent), Urdu (native).